Testing the decay inequality

Clearer plots and a longer, larger-domain experiment

Research question. Does the weighted solution obey the paper's proposed power-law bound across different initial fields? The earlier optical-envelope plots showed spreading. This experiment measures the quantity in the inequality.

We work directly in the canonical coordinates of (6.2), using

$$S(t)f = \mathcal{F}^{-1}(e^{itw}\widehat{f}), \qquad w = k_x^3 + 3sk_x(k_y^2 + k_z^2) + 3\beta(k_y^2 + k_z^2).$$

The two cases are $(s, \beta) = (-1, 0)$ and $(+1, 1)$, matching the minus homogeneous and plus inhomogeneous symbols used in the proofs of Propositions 6.2–6.3. We follow those proofs' positive phase convention; equation (6.1) prints inconsistent signs. No transformation from the original optical coordinates is needed here.

For $\alpha = 1/2$ and $p = \infty$, the diagnostic is

$$R(t; f) = \frac{t^{7/6}\left\| |D_x + \beta|^{1/2} S(t)f \right\|_\infty}{\|f\|_1}.$$

The inequality asserts $R(t; f) \le C$ with C independent of t and f at fixed parameters. The multiplier is $|k_x + \beta|^{1/2}$, not an ordinary spatial derivative. Inputs are normalized to discrete $L^1 = 1$ on each grid.

How to read the plots. A flat ratio corresponds to weighted decay like $t^{-7/6}$ over that interval. A falling ratio indicates faster decay there; a rising ratio indicates slower decay there. **A rising but bounded ratio is compatible with the theorem.** No horizontal "failure threshold" is drawn because the theorem's constant is unspecified. Finite curves cannot establish unbounded growth.

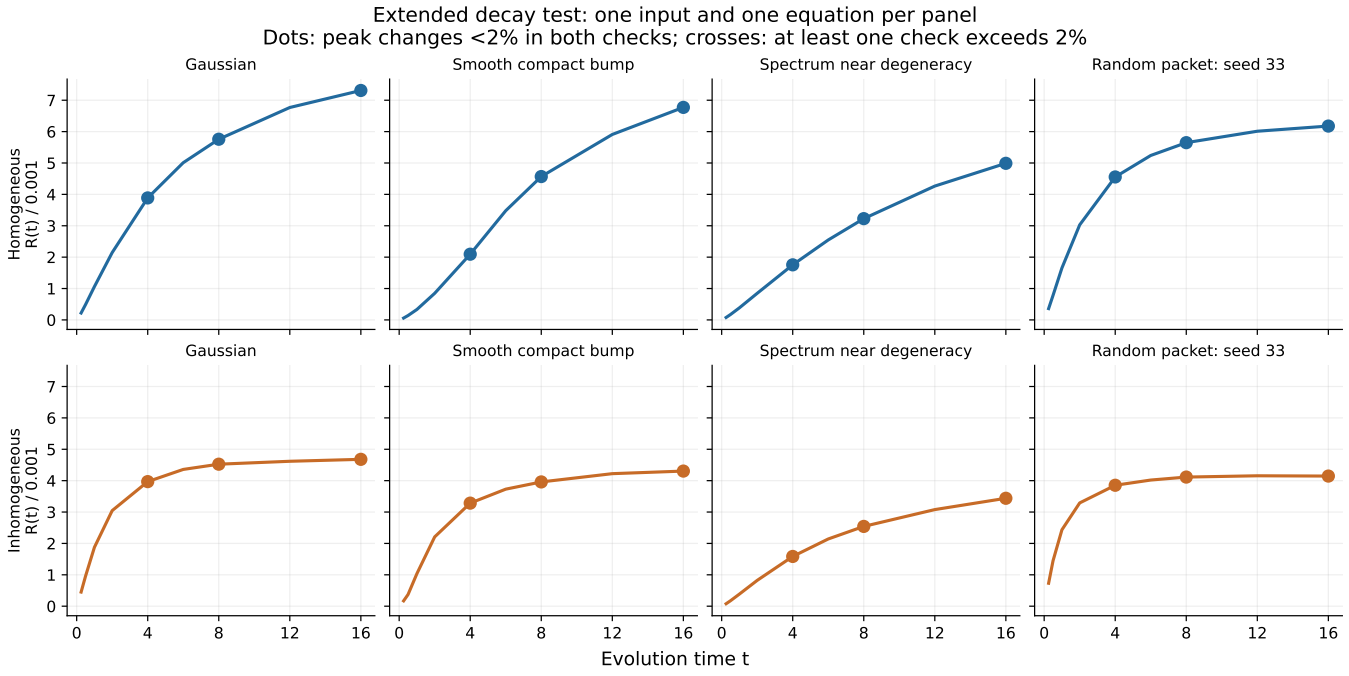


Figure 1. Each panel contains one family and one equation. Curves use a length-288 periodic cube at spacing 1; vertical values are shown in units of 0.001. At $t = 4, 8, 16$, dots indicate less than 2% peak change in both tested refinements; crosses indicate at least one larger change. Intermediate times are not independently checked. The 2% rule is a screening criterion, not a rigorous error bound.

The broader collection of artificial inputs

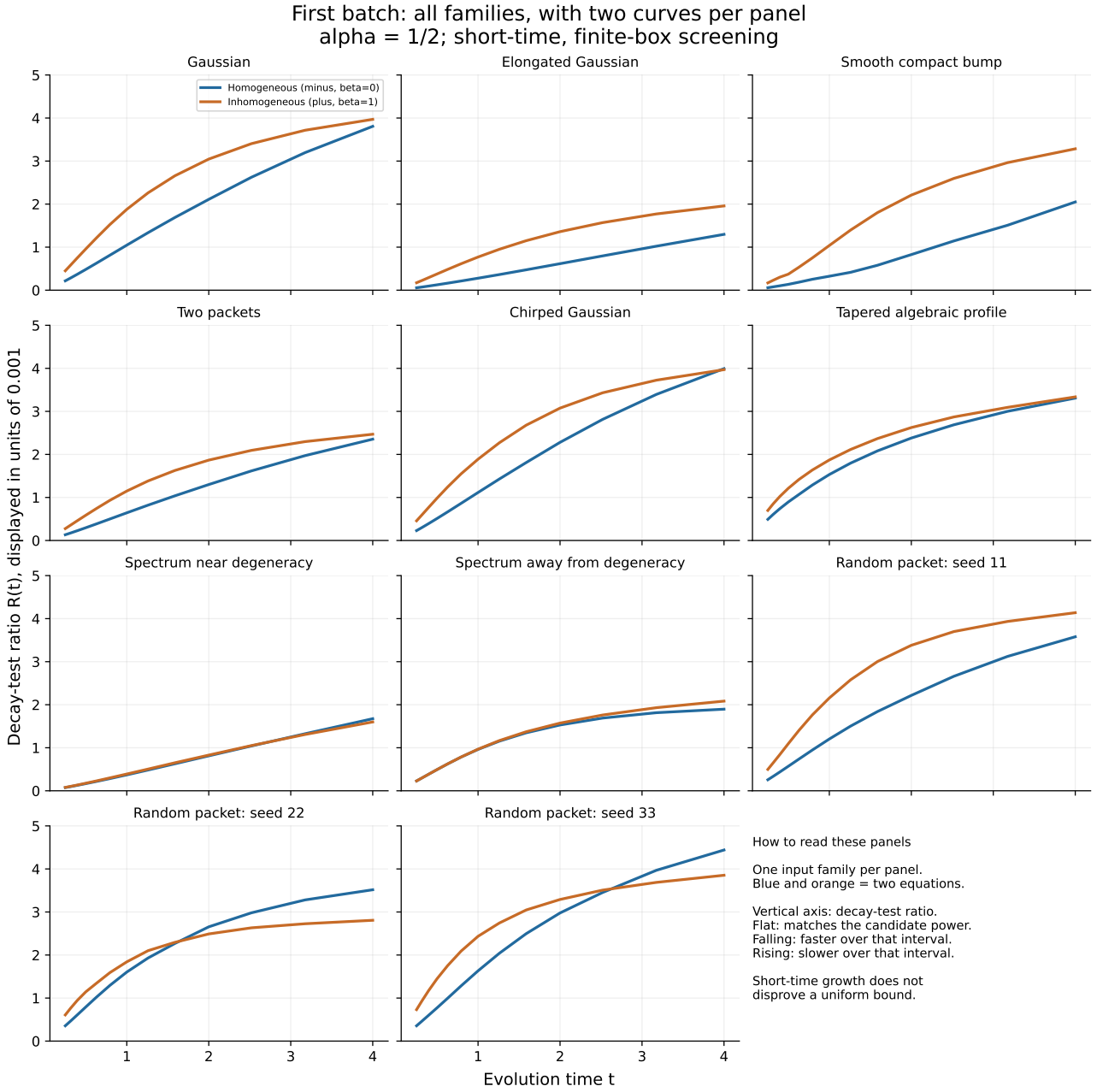


Figure 2. Eleven input families, with two equation variants per panel. These are the original short-time results on a length-96 box, replotted without eleven overlapping curves. Their domain sensitivity remains as recorded in the first batch; the new layout does not make them converged whole-space results.

All recipes define continuous functions that can be resampled on a different grid. The collection includes Gaussian and anisotropic packets, a smooth compact bump, two packets, a chirped phase, a tapered algebraic profile, spectral centers near and away from $k_x = -\beta$, and three seeded random packets. Near/away-degeneracy inputs depend on β and therefore differ between the two equation variants. Random fields combine twelve modes with fixed seeds and a Gaussian spatial window.

What was refined, and what the results support

We extend four families in both regimes to $t = 16$, four times the first batch's maximum time. Gaussian is the reference, the compact bump has a broader spectrum, the near-degeneracy packet targets the multiplier's cusp, and seed 33 represents the random collection. This is a targeted follow-up, not an exhaustive data search.

Domain checks compare lengths 96, 192, and 288 at spacing 1. A separate length-192 grid with spacing 2/3 checks resolution. Every comparison uses the same continuum recipe, normalized using its own grid quadrature. The table gives results at $t = 16$; machine-readable files also include $t = 4, 8$ and complex-field errors on a common cube.

Equation	Input	$R(16)$	Domain peak change	Grid peak change
Homogeneous	Gaussian	0.00731	1.11%	0.18%
Homogeneous	Compact bump	0.00677	0.43%	0.03%
Homogeneous	Near degeneracy	0.00499	1.62%	0.24%
Homogeneous	Random (33)	0.00618	1.61%	0.29%
Inhomogeneous	Gaussian	0.00468	0.01%	0.05%
Inhomogeneous	Compact bump	0.00431	0.00%	0.54%
Inhomogeneous	Near degeneracy	0.00344	1.36%	0.01%
Inhomogeneous	Random (33)	0.00415	0.00%	0.04%

Domain peak change compares lengths 192 and 288; grid peak change compares spacings 1 and 2/3 at length 192. Peak comparisons use each grid's full sampled maximum, not a fixed spatial point. The plotted length-288 field is not itself also refined to spacing 2/3, so the checks do not establish joint convergence.

Observed behavior. The inhomogeneous Gaussian and random packet ratios begin to flatten: from $t = 8$ to 16 they grow by 3.42% and 0.78%, respectively. The near-degeneracy ratios continue to rise more visibly, as do several homogeneous cases. This suggests different transient time scales; it does not establish their eventual behavior.

Numerical qualification. Across all 24 checked case–time combinations, the largest peak changes are 1.62% under domain enlargement and 0.94% under grid refinement. All pass the exploratory 2% screen.The fractional multiplier can produce long spatial tails, and errors can vary nonmonotonically with box size near its cusp. A field's relative L^2 difference and its maximum-amplitude difference are distinct: this follow-up prioritizes the maximum because it appears in the inequality, while retaining both diagnostics.

Inference. These experiments do not establish a faster universal decay rate, a theorem constant, or a counterexample. A convincing candidate violation would require a controlled sequence of times or normalized inputs with increasing ratios that survives numerical refinement. A short transient slope is insufficient. Cases that fail the peak checks should be refined before being interpreted as evidence about the whole-space bound.

Data and reproducibility. Run `python3 decay_dataset.py`, then `python3 refine_decay.py`. Results are in `results/decay_dataset/`: `refined_decay.csv`, `refinement_checks.json`, `REFINEMENT.md`, and the plotted figures. The first batch's inputs remain available as complex-valued NPZ arrays. These are artificial inputs. CDIP buoy observations have been identified as a possible source, but are not imported or presented as three-dimensional measurements.

Source. R. Mosincat, D. Pilod, and J.-C. Saut, *On Third Order Nonlinear Schrödinger Equations*, supplied manuscript `Dysthe3D-2.pdf`, Section 6. This supplement follows the stated estimates as research targets, without claiming to resolve the manuscript's proof details.

Extension through $t = 256$

We now evaluate $t = 16, 32, 64, 128, 256$ for the Gaussian and near-degeneracy packet, in both canonical equation regimes, with $\alpha = 1/2$. These two profiles admit an efficient whole-space transverse integration. Other artificial families have not yet been extended to 256 by this method.

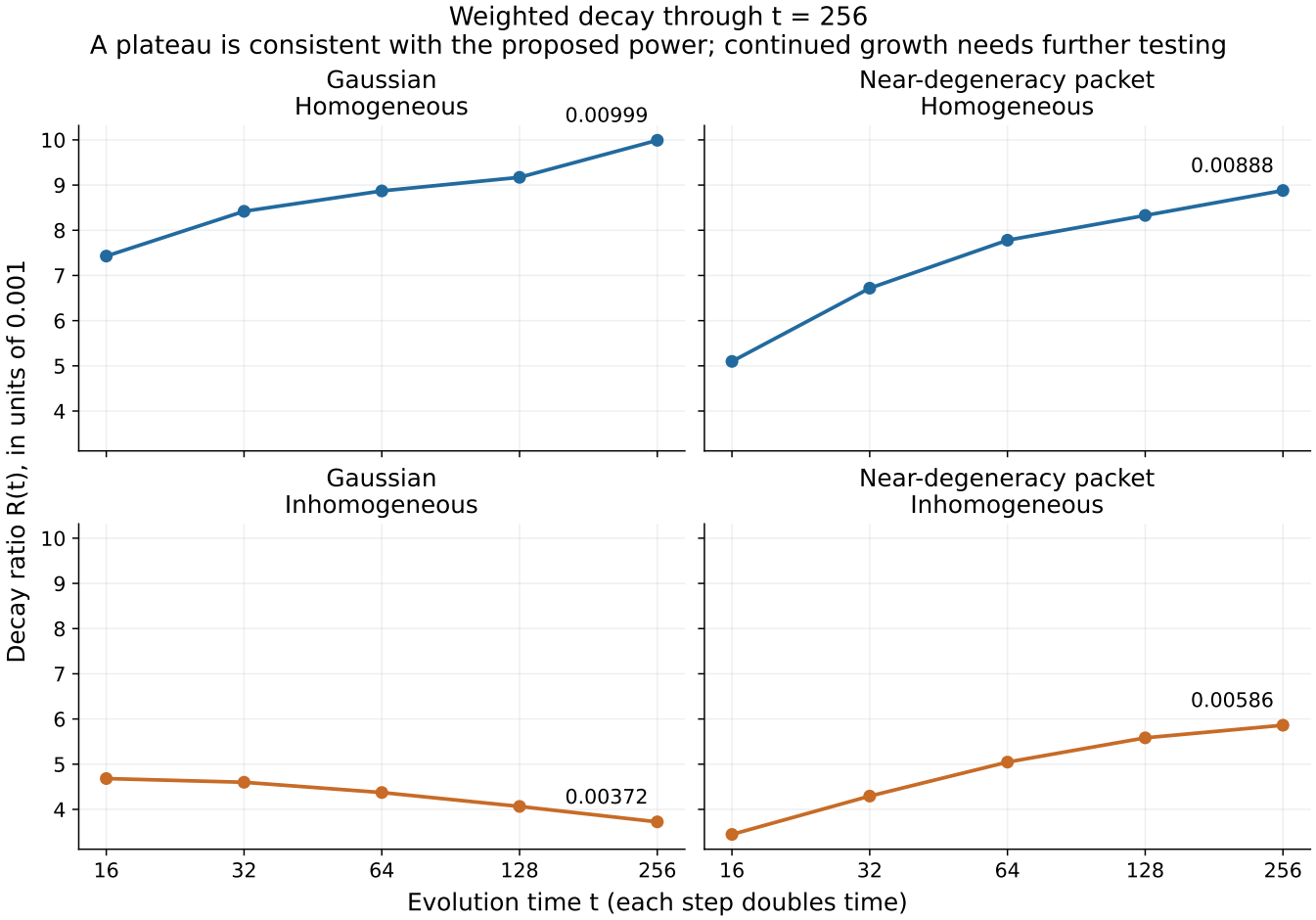


Figure 3. One curve per panel, with time doubling at each marker. The ordinate is the same ratio $R(t)$ as before, displayed in units of 0.001; endpoint labels give its unscaled value. A downward curve means faster weighted decay than $t^{-7/6}$ over that interval. An upward curve can still approach a finite limit and does not by itself contradict the inequality.

Equation	Input	$R(128)$	$R(256)$	Effective exponent
Homogeneous	Gaussian	0.00917	0.00999	1.043
Homogeneous	Near degeneracy	0.00833	0.00888	1.074
Inhomogeneous	Gaussian	0.00407	0.00372	1.293
Inhomogeneous	Near degeneracy	0.00558	0.00586	1.096

For weighted maximum $M(t)$, the last column is $\gamma_{128,256} = -\log[M(256)/M(128)]/\log 2$. It describes only the last time doubling; it is not a fitted asymptotic exponent or a new uniform bound. The paper's comparison exponent at $\alpha = 1/2$ is $7/6 \simeq 1.167$.

Finding. The inhomogeneous Gaussian ratio decreases at late times. The other three ratios increase over the last doubling, by different amounts. None of these finite curves establishes a counterexample. The next question is whether the rising ratios approach a finite limit or continue growing through successive, independently checked time doublings.

How the long-time calculation is checked

Let $r = \sqrt{y^2 + z^2}$ and use the exactly L^1-normalized initial profile

$$f(x, r) = \frac{\exp[-x^2/(2\sigma_x^2) - r^2/(2\sigma_y^2)]e^{iqx}}{(2\pi)^{3/2}\sigma_x\sigma_y^2}.$$

The Gaussian has $(\sigma_x, \sigma_y, q) = (3, 3, 0)$; the near-degeneracy packet has $(6, 3, -\beta)$. Analytically integrating both transverse Fourier variables reduces the weighted field to

$$W(x, r, t) = \frac{1}{4\pi^2}\int_{\mathbb{R}} \frac{|k+\beta|^{1/2}}{d(k,t)} \exp\left[-\frac{\sigma_x^2(k-q)^2}{2} + i(tk^3 + kx) - \frac{r^2}{2d(k,t)}\right] dk, \qquad d(k,t) = \sigma_y^2 - 6it(sk + \beta).$$

This is the same three-coordinate linear solution for these inputs. Only the evaluation method changes. The transverse directions are treated on the whole space; a longitudinal FFT quadrature evaluates many x locations efficiently. Radial symmetry reduces the spatial peak search to (x, r).

Check	x period	Δx	Max. r	Δr
Base	4096	0.5	512	4
Expanded	8192	0.5	1024	4
Refined	8192	0.25	1024	2
Extra expansion at $t = 256$	32768	0.25	2048	2

The full (x, r) search is followed by local optimization of the peak. At 256, the extra expansion changes the sampled peak by at most 0.141%. An independent adaptive quadrature of the integral above agrees with the complex field at each final candidate peak to within 0.020% relative error. The largest sampled search-boundary amplitude is 0.586% of its case's peak. These checks are numerical evidence, not a certification that no larger unsampled peak exists.

Why the previous box was inadequate. At $t = 256$, the inhomogeneous near-degeneracy maximum lies around $x = -841$. It would be outside even the length-288 box used for the earlier $t \le 16$ runs. Extending time without changing the spatial computation would therefore be unreliable.

Remaining limits. The Gaussian-specific reduction cannot be applied unchanged to arbitrary measured fields or every artificial recipe. The adaptive quadrature checks selected peak locations; the global search remains finite and sampled. The two-dimensional radial search, local optimizer status, boundary ratios, and domain/resolution comparisons are all recorded to make these limits reviewable.

Reproduce. Run `python3 long_time_decay.py`, followed by `python3 report/make_long_time_tables.py`. Results and the graph are under `results/long_time/`; compile `report/decay_refinement.tex` for this updated supplement. Source: the same manuscript, canonical symbols (6.2) and weighted estimates in Propositions 6.2–6.3.

Stronger spatial reliability at $t = 256$

Purpose. Hold time, input, and $\alpha = 1/2$ fixed, and vary the spatial computation. This audit separates longitudinal enlargement from transverse enlargement. It tests whether the four reported candidate maxima stabilize to an exploratory relative tolerance of 0.1%.

The longitudinal period increases from 32,768 to 65,536 while the radial limit stays at 2,048. We then search the additional annulus out to radius 4,096. Finally, we double the longitudinal period again to 131,072, retaining the larger radial limit. Every broad scan keeps $\Delta x = 0.25$ and $\Delta r = 2$. The baseline is the saved final scan from the preceding experiment.

Equation	Input	First L change	Radius change	Second L change	Integral check
Homogeneous	Gaussian	0.0130%	0.0000%	0.0046%	0.0025%
Homogeneous	Near degeneracy	0.0020%	0.0000%	0.0007%	0.0003%
Inhomogeneous	Gaussian	0.0001%	0.0000%	0.0000%	0.0000%
Inhomogeneous	Near degeneracy	0.0033%	0.0000%	0.0026%	0.0004%

The three domain columns compare sampled peak magnitudes before and after the stated expansion. The integral check compares the final Fourier-quadrature field to an independently evaluated non-periodic integral at the refined candidate peak. Percentages use the newer or independent result as the denominator. Values are rounded.

Result. 4 of four cases pass the 0.1% peak screen. This includes the two longitudinal expansions, the transverse expansion, the final locally optimized peak drift, and the independent integral check. The largest change in locally optimized peak magnitude between the final two periods is 0.00459%.

Equation	Input	Refined $R(256)$	x	r	Location drift
Homogeneous	Gaussian	0.0099887	-20.7469	0.0000	0.00000
Homogeneous	Near degeneracy	0.0088803	7.7961	6.5381	0.00079
Inhomogeneous	Gaussian	0.0037238	-8.9154	0.0000	0.00002
Inhomogeneous	Near degeneracy	0.0058621	-841.2464	0.0000	0.00021

Locations are continuous-coordinate candidate maxima from the independent integral optimization. The drift is the Euclidean distance in (x, r) between local peak optimizations at the final two longitudinal periods. Its maximum is 0.00079. Axial symmetry means a nonzero r represents a ring of equivalent points, not a unique three-dimensional location.

Additional checks. Local optimization starts from the broad search's largest sample and refines both x and r. Increasing the independent integral's Gauss–Legendre order from 256 to 512 per panel changes the complex field by at most 3.67×10^{-12}% relative to the tighter result. Its integration interval is split at the fractional multiplier cusp and into 64 Gaussian-frequency panels. On each side, $k = -\beta \pm u^2$ removes the square-root cusp from the integration variable. Fixed-order quadrature with an order-doubling check replaces adaptive quadrature, which issued convergence warnings during local searches. The largest sampled search-boundary amplitude is 0.00516% of the corresponding peak. Optimizer status and all intermediate measurements are saved in the per-case JSON records.

What we can now say. At this time and for these four inputs, the peak values have stronger numerical support from separate domain enlargements, continuous local refinement, stable locations, and independent integration. This improves confidence in the reported decay ratios. It does not prove the maximum is global over all of $\mathbb{R}^3$, establish an asymptotic decay rate, or extend reliability to other inputs, values of α, or later times.

Remaining spatial limitation. The broad radial mesh is still spaced by 2; refinement is local to the identified maximum, not a global halving of every cell. A missed narrow peak or a remote maximum is not excluded by these checks. The report therefore claims numerical spatial stability of the identified peaks, not a certified supremum.

Reproduce. Run `python3 spatial_reliability.py`, then `python3 report/make_spatial_tables.py`. Results, configurations, and checkpoints are under `results/spatial_reliability/`.